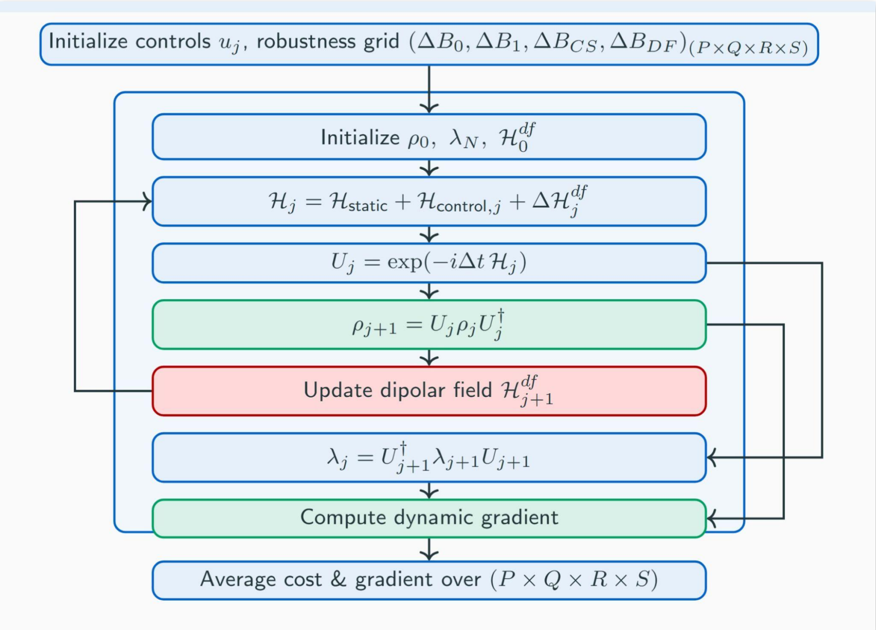
State-Dependent GRAPE: Gradient reformulation for PHIP-SAH polarization transfer in the presence of dipolar field

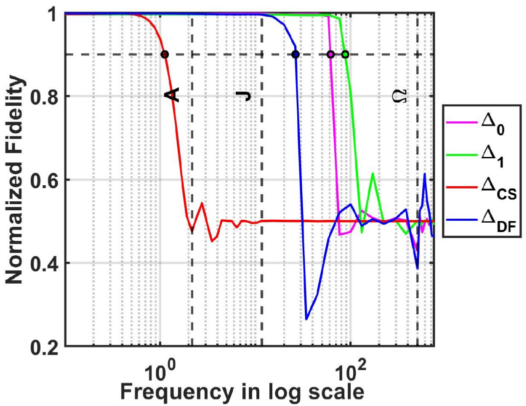
Sreya Das, Dennis Huber, Martin Korzeczek, Martin Planio, Steffen Glaser

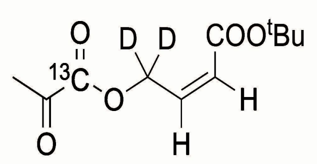
- **Parahydrogen Included Polarization via Side Arm Hydrogenation (PHIP-SAH)** transfers polarization from singlet-order parahydrogen to target ^{13}C nuclei can lead to a **dipolar field build-up over the sample volume, which reduces transfer efficiency and becomes a major limitation for achieving high polarization.**
- Achieving a pulse sequence that is **simultaneously robust against all relevant perturbations — namely B_0 and B_1 - inhomogeneities, chemical shift variations, and dipolar field interaction** remains a significant challenge and an active area of research in this field.

Reformulation of the gradient of the GRAPE algorithm by consistently **incorporating the state-dependent dipolar-field Hamiltonian** into the state and costate evolution, enabling optimal control design for nonlinear spin dynamics.



tert-butyl (Z)-4-((2-oxopropanoyl-1-^{13}C)oxy)but-2-enoate-4,4-d_2





Δ_0 = 61.62 Hz
Δ_1 = 87.98 Hz
Δ_{CS} = 1.11 Hz
Δ_{DF} = 26.39 Hz